

Vaghesan Sundaram

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EDUCATION

University of Maryland, College Park

College Park, MD

B.S. Computer Science (Cybersecurity), B.S. Mathematics, ACES Cybersecurity Minor

Expected December 2027

Relevant Coursework: Data Structures, Algorithms, Computer Systems, Discrete Structures, Linear Algebra, Statistics, Cryptography, Network Security, Penetration Testing, Technical Writing, Programming Languages, Computer Networks, Compilers, Combinatorics and Graph Theory, Number Theory

Middlesex College

Edison, NJ

A.S. Mathematics, completed during high school

June 2024

EXPERIENCE

Cybersecurity Intern

June 2026-Present

Redmind Technologies

Remote

- Identified and documented **7 reproducible findings** across authentication, data exposure, configuration, and dependency risk by combining Semgrep SAST with manual secure code review of a public-facing client application.
- Delivered a developer-facing report and slide briefing that validated evidence, prioritized all 7 findings by severity, and mapped each issue to actionable remediation guidance for developer implementation and client review.
- Developing a PyTorch pipeline with **MuJoCo, robosuite, robomimic, and LeRobot** to measure **training-data and reward-label poisoning** across behavior cloning, transformer, VLA, and offline RL policies.

Information Technology Intern

June 2025-August 2025

New Jersey Courts

Toms River, NJ

- Installed **courtroom recording hardware** to capture legal proceedings and support court recordkeeping.
- Improved IT asset inventory accuracy across **5 departments** (Finance, IT, Civil, Probation, and Operations) by cataloging devices and reconciling physical equipment with tracking records for asset control.

PROJECTS

Agent Least Privilege | *Python, MCP, Pydantic, asyncio, pytest, mypy, Ruff, GitHub Actions*

[GitHub](#)

- Built a Python **MCP proxy** enforcing launch-time policies for filesystem paths, exact arguments, and call quotas.
- Serialized call quotas, recorded redacted audit events, failed closed on errors, and tracked unknown outcomes.
- Blocked **12/12 unauthorized proposals**; completed **10/12 attack tasks** and **24/24 controls** in simulation.

CTI Claim Provenance | *Python, CISA KEV/CSAF, NVD, CVE, MITRE ATT&CK, Pydantic*

[GitHub](#)

- Built a **64-question, 24-group point-in-time CTI benchmark**; constrained output raised evidence binding 83.3% to 86.5% but lowered exact accuracy 33.3% to 26.0%; a 144-response follow-up found no schema benefit.
- Tracked CISA KEV/CSAF, NVD, CVE, MITRE ATT&CK, and vendor source provenance with SHA-256 hashes.
- Enforced **point-in-time cutoff and source-authority rules** with evidence spans, abstention, and human review.
- Automated typed-schema validation, split isolation, abstention, secret scanning, and privacy-safe aggregate checks.

Keystroke Authentication System | *Python, Flask, SQLite, JWT, bcrypt, CSRF, CORS*

[GitHub](#)

- Secured 2 Flask services using **3-sample Pearson/MSE keystroke profiles**, 20-character bcrypt fallback, 5-attempt/15-minute rate limiting, 60-second single-use codes, 1-hour JWTs, and atomic profile re-enrollment.
- Hardened login via CSRF, CORS allowlisting, JWT binding, **callback validation**, and server-to-server exchange.

Prophet Prediction Market | *Python, WebSockets, sentence embeddings, clustering*

[GitHub](#)

- Contributed to an autonomous prediction-market platform integrating **xAI Grok event detection**, OpenRouter trading agents, and Solana smart contracts for on-chain market creation and trading at HackPrinceton.
- Built a 4-route Flask pipeline processing up to **40 X posts/min** by weighting 7 signals and clustering 10+ posts.

TECHNICAL SKILLS

Certification: CompTIA Security+ (SY0-701), March 2026-March 2029

Languages: Python, Java, C++, C, Assembly, Rust

Security: OSINT, Maltego, Shodan, Recon-ng, Maigret, Burp Suite, Nmap, Wireshark, Ghidra, vulnerability triage, TCP/IP, DNS, DHCP, PCAP analysis, Kali Linux, Hashcat, John the Ripper, privilege escalation, exploit development

Tools: Git, GitHub Actions, Docker, Linux, PowerShell, Flask, Pydantic, pytest, Hypothesis, mypy, REST APIs, GDB

Data & ML: PyTorch, MuJoCo, robosuite, robomimic, LeRobot, SQL, SQLite, NumPy, SciPy, scikit-learn